Myungsin Cho

Teaching Statement

I have taught mathematics at different levels and in a range of formats. My experience spans general-education courses for non-majors, service courses such as calculus, proof-based linear algebra, and applied mathematics for STEM students. I have taught in traditional classrooms and large lectures with more than 100 students, as well as in flipped classrooms and synchronous online lectures.

Across these settings, my goal remains the same: to help students develop the ability to think mathematically for themselves. I approach mathematics as one connected story, organizing courses around questions that motivate new concepts and drawing attention to patterns of reasoning that recur across different subjects.

Teaching Mathematics as a Connected Whole

A logically organized course does not always make the connections between its topics apparent to students. I therefore consider not only what mathematics to teach, but also when and how to introduce new ideas so that their motivations become clear.

In Calculus I, students can easily perceive limits, derivatives, and integrals as separate topics, each with its own techniques and procedures. To address this, I organized the course around approximation as a unifying theme. Throughout the semester, I repeatedly returned to this perspective, helping students recognize how seemingly different concepts develop from a common mathematical idea.

In proof-based Linear Algebra, I structured the course so that questions introduced early in the semester would be revisited as students developed a deeper understanding of the subject. We began with matrices and systems of linear equations, learning concrete methods of computation while leaving some fundamental questions unresolved. As students encountered more abstract concepts such as vector spaces and linear maps, we returned to these earlier questions from a new perspective. For example, the seemingly arbitrary definition of matrix multiplication became natural through the composition of linear maps, and results initially proved using Gaussian elimination could be understood through more structural arguments. In this way, students could revisit familiar mathematics and discover how abstract ideas reveal the underlying connections.

The connections I emphasize extend beyond mathematical concepts to the questions we ask and the proofs we construct. Questions of existence, uniqueness, and classification recur across different subjects, even when the objects under study change. Likewise, proofs of seemingly different results often share a common logical structure. When reading a textbook proof, I ask students to identify its central idea and consider how the author might have discovered the argument before working through its details.

Supporting Students' Learning

Different courses have different educational purposes, and I adapt my teaching accordingly. In Math of Beauty and Decision, a general-education course for nonmajors, I used examples involving voting, symmetry, and games to help students explore familiar situations from a mathematical perspective. In courses for STEM students, I place greater emphasis on building the mathematical foundations and reasoning skills they will need for further study, including understanding when and why to apply particular mathematical methods.

I also value building relationships with my students that extend beyond the immediate demands of a course. I enjoy learning about their interests and discussing how mathematics relates to their studies and future goals. Having studied engineering as an undergraduate, I draw on my own experience to share concrete examples of how mathematics is used in other fields. Through these conversations, I have also advised students on which mathematics courses to take next and on their plans for graduate study.

Students within the same course also have different levels of preparation and mathematical maturity. When someone has trouble following an argument, I ask where the reasoning stopped being clear and work

from that point, often adding a more concrete example. For students seeking greater challenges, my lecture notes include material beyond what we cover in class, along with occasional questions in footnotes that invite further exploration.

In Analysis and Optimization, I have reordered some topics to introduce new concepts closer to the problems that motivate them. When I noticed that students with little experience writing proofs struggled to get started on assignments, I redesigned some homework problems as guided worksheets. These provide the overall structure of an argument while leaving key steps for students to develop and justify themselves. Students reported that the worksheets made proofs more accessible and intuitive.

I design assessments to reward consistent practice of fundamental skills while maintaining high expectations for conceptual understanding and mathematical reasoning. In Calculus, I allowed students to revisit missed exam problems for partial credit, giving them an opportunity to learn from their mistakes. As AI has made take-home work less reliable as a measure of independent understanding, I have placed greater emphasis on regular supervised assessments. In my current Analysis and Optimization course, homework provides low-stakes practice, and weekly quizzes draw directly on assigned homework problems. Quiz performance can count toward the computational part of the corresponding exam, allowing students to demonstrate mastery of fundamental skills through regular practice while exams devote greater attention to conceptual and proof questions.

Fostering Independent Mathematical Thinking

Explaining mathematics to another person is an important part of understanding it. In Linear Algebra, I used an oral final examination and provided sample problems and possible follow-up questions in advance. Preparing to explain their reasoning gave students an opportunity to identify what they understood and where they needed further work. During the examination, I could ask them to clarify a step or explain a choice, allowing me to assess their understanding beyond what a written answer alone might reveal.

The growing use of AI makes independent mathematical judgment increasingly important. I emphasize to students that using AI properly and responsibly requires more than effective prompting. They need enough mathematical understanding to examine assumptions, identify gaps in reasoning, and verify conclusions produced by these tools. I generally require students to submit homework by hand. Even when they have consulted external tools, writing out a solution gives them another opportunity to work through and verify the reasoning before submitting it.

My experience mentoring through a Directed Reading Program has also shaped how I think about undergraduate research. I encouraged students to raise their own questions and explain the material they studied, rather than simply complete readings and exercises. In undergraduate research, I would build on this approach by starting with accessible mathematical examples and helping students develop their observations into questions they can investigate. As a project progresses, I would give students greater responsibility for deciding what to explore next. Discovering new results would be rewarding, but I believe that learning to formulate and pursue mathematical questions is itself a valuable research experience.